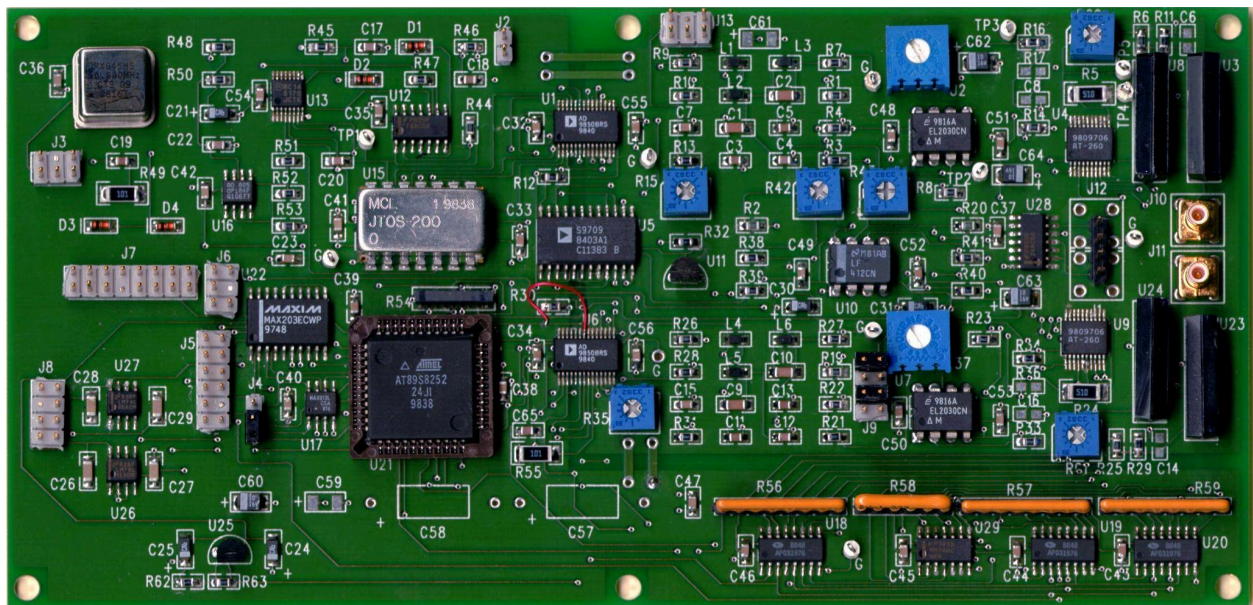


Input image 1 (pcbclear2.jpg):



Output of [main.py](#) for image 1:

```
--- LLM Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'General Purpose', 'estimated_bom_inr': '175-2050 INR', 'reasoning': 'Based on the component count of 350, which is a high number, and diverse types including capacitors, resistors, and ICs. The coverage suggests a general-purpose PCB with no specific designations indicating a highly specialized board.'}}

--- Reflection Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'General Purpose', 'estimated_bom_inr': '175-2050 INR', 'reasoning': 'Based on the component count of 350, diverse types including capacitors, resistors, and ICs. The OCR detected 4 ICs but CV did not detect any, suggesting a general-purpose PCB with no specific designations.'}}
{'status': 'success', 'result': {'complexity': 'High', 'pcb_type': 'General Purpose', 'estimated_bom_inr': '175-2050 INR', 'reasoning': 'Based on the component count of 350, diverse types including capacitors, resistors, and ICs. The OCR detected 4 ICs but CV did not detect any, suggesting a general-purpose PCB with no specific designations.'}, 'steps_used': 4}
(venv_clean)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version5 (main)
$
```

Output of [app.py](#) for image 1:

localhost:8501

Deploy

Chat

Analysis

Time taken: 289.27 seconds

Agent completed in 4 step(s)

BOM Estimation Result

```
{
  "complexity": "High"
  "pcb_type": "Complex Digital PCB"
  "estimated_bom_inr": "210-850 INR"
  "reasoning":
    "The component count is high with a diverse mix of components, including 39 capacitors and 17 ICs. The presence of connectors suggests a digital or mixed-signal application. OCR provided potential IC names but their actual functionality remains uncertain."
}
```

Input image 2 (pcbimagetrial4k.png):



Output of [main.py](#) for image 2:

```
--- LLM Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'General-purpose PCB', 'estimated_bom_inr': '420-21000 INR', 'reasoning': 'The component count is high at 850, with a significant number of unidentified components (808 unknown). The presence of ICs suggests functionality beyond basic circuits. Given the variability in component types and counts, the BOM cost ranges widely.'}}

--- Reflection Response ---
{'final_answer': {'complexity': 'High', 'pcb_type': 'General-purpose PCB', 'estimated_bom_inr': '420-5000 INR', 'reasoning': 'Component count is high at 850, with a significant number of unknown components (808). Given the presence of ICs and variability in component types, the BOM cost ranges widely.'}}
{'status': 'success', 'result': {'complexity': 'High', 'pcb_type': 'General-purpose PCB', 'estimated_bom_inr': '420-5000 INR', 'reasoning': 'Component count is high at 850, with a significant number of unknown components (808). Given the presence of ICs and variability in component types, the BOM cost ranges widely.'}, 'steps_used': 4}
(venv_clean)
HP@DESKTOP-LNORTVI MINGW64 ~/project_python/BOMCalculation/version5 (main)
$
```

Output of [app.py](#) for image 2:

localhost:8501
Refresh (Ctrl+R)

Deploy

Analysis

Time taken: 280.74 seconds

Agent completed in 4 step(s)

BOM Estimation Result

```
{
  "complexity": "High"
  "pcb_type": "Unknown"
  "estimated_bom_inr": "425-41250 INR"
  "reasoning":
    "Component count is high with mostly unknown components. IC count detected by CV is low, suggesting a simple board layout despite the component count."
}
```